GHISLAIN FRANK TONTSA APALA

Toulouse, France | frankapala75@gmail.com | linkedin.com/in/tontsa | github.com/FrankApala | ta-embedded.com

Computer and embedded systems engineer (UTBM), currently specializing in systems engineering at ISAE-SUPAERO (Advanced Master, requirements engineering, system architecture, MBSE). Combining hands-on embedded software experience (C/C++, RTOS, Linux) with training in the design of complex systems. Seeking a 1-year apprenticeship in systems engineering, system architecture or embedded software.

EDUCATION

ISAE-SUPAERO / TSAAE — *Toulouse, France*
Advanced Master, Systems Engineering (RNCP Level 7) — *Sept. 2026 – Sept. 2027*

- State-recognized professional certification: Systems Engineering Architect-Manager
- Coursework: requirements engineering, systems architecture and design, MBSE, dependability, project management

Université de Technologie de Belfort-Montbéliard (UTBM) — *Belfort, France*
Engineering Degree, Computer Science — *Sept. 2022 – 2025*

RPTU Kaiserslautern — *Germany*
Exchange Semester (ERASMUS), Computer Science Engineering — *Oct. 2024 – Mar. 2025*

ISET Nabeul — *Tunisia*
Bachelor's Degree, Computer Science: Embedded and Mobile Systems — *Sept. 2019 – Jun. 2022*

PROFESSIONAL EXPERIENCE

EFI Automotive — *Beynost, France*
Embedded Software Developer — *Mar. 2025 – Aug. 2025*

- Implemented a UDS server (ISO 14229) over CAN under FreeRTOS on a dsPIC33CK256MP508 microcontroller
- Improved memory management and task scheduling to reduce power consumption and increase system robustness
- Integrated a watchdog mechanism to prevent system lock-ups and secure software execution
- Set up functional tests with Vector CANoe to validate compliance with ISO standards

Oxynode — *Belfort, France*
Freelance IoT Developer — *Aug. 2024 – Nov. 2024*

- Developed JavaScript decoders to process and normalize data from IoT sensors connected via LoRaWAN
- Implemented unit tests and validated incoming data frames, ensuring reliability of processed data

Junior UTBM / ALSTOM — *Belfort, France*
Project Manager — *Mar. 2024 – Aug. 2024*

- Analyzed client needs and drafted requirements specifications for missions assigned by companies to UTBM students
- Tracked schedule and deliverables through regular meetings with the team and client companies
- Led an IoT project, coordinating exchanges between the student team and the client

INMAN — *Haguenau, France*
Embedded Software Developer — *Sept. 2023 – Feb. 2024*

- Ported Zephyr RTOS firmware from an electronic mixer design to proportional valve control
- Designed a PID control loop with sensor feedback, calibration, and safe fallback on fault
- Developed an ESP32-based embedded tool to test and control a hydraulic system via RS232

SKILLS

Languages: C, C++, Python

Systems & Protocols: Linux, FreeRTOS, Zephyr RTOS, CAN, UART, SPI, I²C, MQTT

Hardware architectures: ARM Cortex-A, ARM Cortex-M, PIC 8/16/32-bit, FPGA

Microcontrollers & platforms: STM32, ESP32, Raspberry Pi, PIC/dsPIC

Build & automation: Makefile, CMake, Yocto, Buildroot

Tools & debugging: Git, GDB, MPLAB ICD, Oscilloscope, CANoe

Languages spoken: French (native), English (native)

DISTINCTIONS

- Eiffel Excellence Scholarship, French Ministry for Europe and Foreign Affairs (2022–2025)
- Tunisia–Cameroon Cooperation Scholarship (2019–2022)